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Ch.11 Human Circulatory System

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

1. Compared with a vein, an artery usually has which feature?

Answer: (d) Has no internal valves — Arteries carry blood under high pressure and generally lack valves; veins have valves to prevent backflow.

2. The bicuspid valve guards which opening?

Answer: (d) Left atrium and left ventricle — The mitral/bicuspid valve lies between the left atrium and left ventricle.

3. What is the state of AV valves at the end of the first heart sound?

Answer: (d) Both are closed — The first sound (lubb) is produced by closure of the tricuspid and bicuspid valves.

4. At normal rate, approximately how much blood does the heart pump per minute?

Answer: (c) 5 litres — Cardiac output is about 5 L/min at rest.

5. Closure of tricuspid and bicuspid valves produces which sound?

Answer: (a) “Lubb” — AV-valve closure causes the first heart sound.

6. The SA node initiates the heartbeat in which chamber first?

Answer: (a) Right atrium only — The SA node is located in the right atrium and starts the impulse there before it spreads.

7. Normal systolic pressure in a young adult is about what?

Answer: (d) 120 mm Hg — Typical resting BP is about 120/80 mm Hg.

8. Blood pressure is highest in which vessels, and flow is slowest in which?

Answer: (b) Arteries, capillaries — Pressure is highest in arteries; large total cross-sectional area makes capillary flow slow.

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9. A “lub-hiss” sound instead of normal “lub-dubb” suggests what?

Answer: (c) Defective semilunar valve — A hiss/murmur often indicates valve incompetence or stenosis; the textbook links it to defective semilunar valves.

10. Which system transports materials besides the blood circulatory system?

Answer: (a) Lymphatic system — Lymphatics return tissue fluid, transport fats and support immunity.

SHORT QUESTIONS

1. What is the main difference between artery and vein walls?

Arteries have thick, muscular and elastic walls to withstand high pressure. Veins have thinner walls, larger lumens and valves to prevent backflow.

2. List the four heart valves and their locations.

Valve	Location
Tricuspid	Right atrium → right ventricle.
Bicuspid/mitral	Left atrium → left ventricle.
Pulmonary semilunar	Right ventricle → pulmonary artery.
Aortic semilunar	Left ventricle → aorta.

3. State the phases of the heartbeat.

Atrial systole, ventricular systole, and joint diastole (complete cardiac diastole).

4. List the principles and uses of an electrocardiogram.

An ECG records electrical activity of the heart through skin electrodes. P wave = atrial depolarization, QRS = ventricular depolarization, T = ventricular repolarization. It is used to detect arrhythmias, conduction defects, ischemia/infarction and chamber enlargement.

5. Define angiography and angioplasty.

Angiography is X-ray imaging of blood vessels after injecting contrast dye. Angioplasty is widening a narrowed vessel using a balloon catheter, often with a stent.

6. What are Purkinje fibres?

Purkinje fibres are specialized cardiac conducting fibres in the ventricular walls that rapidly spread the impulse from the bundle branches, causing coordinated ventricular contraction.

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7. What are vasoconstriction and vasodilation?

Vasoconstriction is narrowing of blood vessels by smooth-muscle contraction, raising resistance/pressure. Vasodilation is widening by relaxation, lowering resistance and increasing flow.

8. What is the rate of blood flow in different vessel types?

Velocity is highest in the aorta (about 450–500 mm/s), falls through arteries and arterioles, is lowest in capillaries (about 0.1 mm/s) because of their huge total cross-sectional area, and rises again in venules, veins and venae cavae (about 250–300 mm/s).

9. State the role of baroreceptors and volume receptors in BP regulation.

Baroreceptors in the carotid arteries and aortic arch detect pressure changes and signal the brain to adjust heart rate/force and arteriolar diameter. Volume regulation involves brain receptors that trigger thirst and ADH release (water retention), and stretch receptors in the right atrium that release atrial natriuretic hormone (ANH) to increase salt/water excretion and lower blood volume/pressure.

10. Differentiate between thrombus and embolus.

Thrombus	Embolus
Definition: Clot attached to vessel wall at site of formation.	**Definition:** Detached clot/particle travelling in blood.
Effect: Blocks locally.	**Effect:** Blocks at a distant narrow vessel.

LONG QUESTIONS

1. Describe the structure of heart walls and rationalize chamber thickness.

The heart wall has three layers: epicardium (outer, visceral pericardium), myocardium (middle cardiac muscle) and endocardium (inner endothelial lining). The heart is enclosed in the pericardium, which protects and lubricates it.

- **Atria:** thin walls because they only push blood into the ventricles.
- **Right ventricle:** moderately thick because it pumps blood only to the nearby lungs (low pressure).
- **Left ventricle:** thickest wall (about three times the right ventricle) because it pumps blood to the entire body at high pressure.

Wall thickness is therefore directly related to the pressure each chamber must generate.

2. Describe blood flow through the heart as regulated by valves.

Vena cava → right atrium → Tricuspid valve → right ventricle → Pulmonary semilunar valve → pulmonary artery → lungs → Pulmonary veins → left atrium → Bicuspid/mitral valve → left ventricle → Aortic semilunar valve → aorta → body

During ventricular systole, tricuspid and bicuspid (AV) valves close to prevent backflow into atria, producing the “lubb” sound; pulmonary and aortic semilunar valves open. During ventricular diastole, semilunar valves close to prevent backflow from arteries, producing the “dubb” sound, while AV valves open for filling.

3. Explain how a heartbeat is initiated and controlled.

SA node (pacemaker) fires in right atrium → Impulse spreads over atria → atrial systole → AV node delays impulse (~0.15 s) → Bundle of His → left/right bundle branches → Purkinje fibres in ventricular walls → Ventricles contract almost simultaneously

The SA node sets the normal rhythm (~70 beats/min). The AV node delay allows atria to empty before ventricles contract. The brain modifies rate: sympathetic nerves and adrenaline increase it during exercise/fever; parasympathetic (vagus) slows it at rest/sleep. If the SA node is weak or conduction is blocked, an artificial pacemaker may be used.

4. Describe the detailed structure of arteries, veins and capillaries.

Arteries	Veins	Capillaries
Wall: three tunics; thick tunica media with muscle and elastic fibres.	**Wall:** three tunics; thin tunica media, less muscle/elastic tissue.	**Wall:** single layer of endothelial cells.
Lumen: narrow; stays open when empty.	**Lumen:** wide; collapses when empty; valves prevent backflow.	**Lumen:** very narrow (~8 µm); RBCs pass single file.
Function: carry blood away from heart under high pressure.	**Function:** return blood to heart at low pressure; skeletal muscle pump assists.	**Function:** exchange of gases, nutrients and wastes with tissues.

All arteries carry oxygenated blood except pulmonary arteries; all veins carry deoxygenated blood except pulmonary veins.

5. Describe the role of precapillary sphincters in capillary flow.

Precapillary sphincters are rings of smooth muscle at the entrance of capillary loops from arterioles. They open or close according to the metabolic needs of the tissue.

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When tissue activity rises, O_2 falls and CO_2/metabolites rise, causing vasodilation and sphincter opening, so more blood flows through capillaries. When activity falls, sphincters constrict and reduce flow. This directs blood to active tissues and regulates capillary perfusion.

6. Write the components of pulmonary circulation.

> Right ventricle → Pulmonary trunk → Right/left pulmonary arteries → Pulmonary arterioles → lung capillaries (gas exchange) → Pulmonary venules → pulmonary veins → Left atrium

Pulmonary circulation carries deoxygenated blood to the lungs and returns oxygenated blood to the heart. It operates at lower pressure than systemic circulation to protect delicate alveolar capillaries.

7. What are the main components of coronary, hepatic-portal and renal circulation?

- **Coronary circulation:** right/left coronary arteries arise from the aorta → arterioles → capillaries in heart muscle → coronary veins → coronary sinus → right atrium.
- **Hepatic portal circulation:** capillaries of stomach, intestine, spleen and pancreas → hepatic portal vein → liver sinusoids → hepatic veins → inferior vena cava. The liver processes absorbed nutrients and removes toxins before blood reaches the heart.
- **Renal circulation:** renal artery → smaller arteries → afferent arteriole → glomerulus → efferent arteriole → peritubular capillaries/vasa recta → venules → renal vein.

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8. Define blood pressure and explain systolic and diastolic pressure.

Blood pressure is the force exerted by blood against the inner walls of blood vessels, usually referring to systemic arterial pressure. It is measured with a sphygmomanometer and expressed as systolic/diastolic mm Hg.

- **Systolic pressure:** maximum arterial pressure during ventricular systole; about 120 mm Hg in a normal young adult.
- **Diastolic pressure:** minimum arterial pressure during ventricular diastole; conventionally about 80 mm Hg.

The textbook states diastolic pressure as 86 mm Hg in one line but uses the conventional 120/80 reading elsewhere; 120/80 is the standard reference value.

9. Identify factors causing atherosclerosis and arteriosclerosis.

Atherosclerosis is the deposition of materials within arteries: fatty materials, cholesterol, abnormal smooth muscle cells, fibrin and cellular debris. Cholesterol accumulation is the prime contributor. It narrows arteries, impairs function and increases the risk of thrombus, heart attack and stroke.

Arteriosclerosis is hardening of arterial walls, often due to calcium deposition. The arteries lose elasticity and cannot expand normally, forcing the heart to work harder. Severe atherosclerosis usually leads to arteriosclerosis.

- Risk factors: high LDL/cholesterol, smoking, hypertension, diabetes, obesity, inactivity, age and heredity.

10. Write notes on angina pectoris, heart attack and heart failure.

Condition	Meaning	Key features
Angina pectoris	Chest pain due to insufficient blood/oxygen supply to cardiac muscle, usually from narrowed coronary arteries.	Pain during stress/exertion; warning sign of possible future heart attack.
Heart attack (myocardial infarction)	Death of heart muscle due to complete blockage of coronary blood flow, usually by thrombus on atherosclerotic plaque.	Severe prolonged chest pain, sweating, nausea; emergency.
Heart failure	Heart cannot pump enough blood to meet tissue needs; may follow prolonged hypertension or muscle damage.	Breathlessness, fatigue, oedema; congestive failure may retain blood in heart/lungs.

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11. Outline the main principles of coronary bypass and angioplasty.

Coronary bypass surgery: a healthy blood vessel is taken from the leg, arm, chest or abdomen and grafted above and below a blocked coronary segment, so blood bypasses the blockage. It does not cure atherosclerosis, so lifestyle changes remain essential.

Angioplasty and stenting: a catheter with a balloon is guided into the narrowed artery under X-ray guidance. The balloon is inflated to dilate the artery; an expandable mesh stent may be placed to keep it open. Dye confirms successful dilation.

12. Define hypertension and describe factors regulating BP that can cause hyper/hypotension.

Hypertension is chronic elevation of blood pressure, consistently above 140/90 mm Hg.

- **Regulation:** cardiac output, blood volume, vessel resistance, baroreceptors (carotid/aortic), kidneys and hormones (ADH, aldosterone, adrenaline, ANH).
- **Causes of hypertension:** abnormal nervous/hormonal regulation, stress, obesity, high salt intake, smoking, heredity, kidney disease.

- **Causes of hypotension:** dehydration, blood loss, heart failure, excessive vasodilation or severe infection.

Prolonged hypertension can cause brain haemorrhage, congestive heart failure and kidney damage.

13. List lifestyle changes that protect against hypertension and cardiac problems.

- Stop smoking and limit alcohol.
- Eat a balanced diet low in salt, saturated fat and sugar; increase fruits, vegetables and fibre.
- Exercise regularly and maintain a healthy weight.
- Manage stress and get adequate sleep.
- Control diabetes, cholesterol and blood pressure; take prescribed medicines regularly.
- Avoid excessive caffeine and monitor BP periodically.

14. Describe the structure and role of lymph capillaries, vessels and ducts.

Lymph capillaries are blind-ended, highly permeable vessels in tissues that collect excess interstitial fluid; once inside, the fluid is called lymph. In intestinal villi, specialized lymph capillaries (lacteals) absorb fats.

Lymph capillaries join into larger lymph vessels with valves; flow is maintained by skeletal muscle contraction, visceral movement and breathing. Lymph nodes filter lymph and contain lymphocytes/macrophages that destroy pathogens.

Lymph vessels join into two main ducts: the right lymphatic duct drains into the right subclavian vein, and the thoracic duct drains into the left subclavian vein. About 3 L of excess fluid is returned to blood daily, maintaining blood volume, transporting fats and supporting immunity.

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INQUISITIVE QUESTIONS

1. Why is pulmonary circulation pressure lower than systemic circulation?

The lungs are close to the heart and delicate; high pressure would damage alveolar capillaries and cause oedema. The right ventricle therefore pumps at lower pressure.

2. Why must the heart develop early and function in the embryo?

As the embryo grows, diffusion alone cannot supply oxygen/nutrients or remove wastes. An early beating heart establishes circulation to support rapid growth and organ development.

3. How do vasoconstriction/vasodilation reflect emotions?

Emotions activate the autonomic nervous system. Fear/stress may cause vasoconstriction in skin (pallor) and vasodilation in muscles; embarrassment causes facial vasodilation (blushing).

4. How is the blood circulatory system dependent on the lymphatic system?

Lymphatics return leaked plasma proteins and fluid to blood, maintain blood volume, absorb dietary fats, and filter pathogens. Without them, oedema and immune/fat-transport problems would occur.

5. Why is swollen lymph nodes a cause of concern?

Swelling indicates active immune response to infection, inflammation or sometimes malignancy. Persistent, hard or generalized enlargement needs medical evaluation.

6. Trace lymph from a lymph capillary until it returns to blood.

> Lymph capillary → Lymph vessel → Lymph node → Larger lymph trunk → Thoracic duct/right lymphatic duct → Subclavian vein

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